

33-2 Cisco Device Security Troubleshooting - Answer Key

In this lab you will troubleshoot device security for a Cisco router.

Cisco Device Security Troubleshooting

A junior network administrator was tasked with configuring device security on R1 as follows:

- 1) Only allow remote access to the command line from your administrator workstation PC1. No other PCs on the 10.0.0.0/24 subnet or anywhere else in the network are allowed to access the router.
- 2) Only SSH version 2 is allowed for remote management access.
- 3) Login with the username **admin** and the password **Flackbox1**.
- 4) Must have access to global configuration mode.
- 5) Must not configure **any** other passwords.

You have arrived in the office but are unable to SSH to R1 from PC1.

Your task is to troubleshoot and correct the configuration.

You must discover what has been misconfigured so the junior engineer can be retrained. Verify all problems before correcting the configuration.

There is not a set order of actions to troubleshoot this scenario. Troubleshooting in a logical fashion will however make it easier and quicker. This is how I would do it.

Verify you cannot SSH to R1 from PC1.

```
C:\>ssh -l admin 10.0.0.1
```

```
% Connection refused by remote host
```

Verify you have IP connectivity.

```
C:\>ping 10.0.0.1
```

Pinging 10.0.0.1 with 32 bytes of data:

```
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 10.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Use a console connection to check the virtual terminal lines on R1.

```
R1#show run
Building configuration...
```

```
!
line vty 0 4
access-class 1 in
login local
line vty 5 15
access-class 1 in
login local
!
truncated
```

We can see an inbound access list has been enabled on the virtual terminal lines. Check the access list configuration.

```
R1#show access-lists 1
Standard IP access list 1
deny 10.0.0.0 0.255.255.255 (48 match(es))
permit host 10.0.0.10
```

The first line in the ACL denies traffic from all hosts on the 10.0.0.0/24 network, including PC1 at 10.0.0.10. SSH traffic from PC1 never reaches the 2nd line of the ACL to be permitted.

There is no need for the deny 10.0.0.0 0.255.255.255 entry in the ACL because there is an implicit deny any at the end of access lists. Correct the ACL.

```
R1(config)#no access-list 1
R1(config)#access-list 1 permit host 10.0.0.10
```

It is important to re-check the configuration to ensure incoming management traffic will only be allowed from PC1.

```
R1#show access-lists 1
Standard IP access list 1
permit host 10.0.0.10
```

```
R1#show run
Building configuration...
!
line vty 0 4
access-class 1 in
login local
line vty 5 15
access-class 1 in
login local
!
truncated
```

Try the SSH connection from PC1 again.

```
C:\>ssh -l admin 10.0.0.1
Open
Password:
R1>
```

Incoming SSH connections from PC1 are accepted now. Verify SSH from PC2 is rejected.

```
C:\>ssh -l admin 10.0.0.1
```

```
% Connection refused by remote host
```

Back on PC1, verify you have access to global configuration mode.

```
R1>enable
% No password set.
```

There is no enable secret so we cannot get to global configuration mode. We have been forbidden from configuring an enable or any other password so we need another way to give the admin user access.

Over the console connection, set privilege level 15 for the admin username. This will give access at the enable prompt level upon login.

```
R1(config)#username admin privilege 15 secret Flackbox1
```

Back on PC1, open a new SSH session and verify you can get to global configuration mode now.

```
R1>exit
```

```
[Connection to 10.0.0.1 closed by foreign host]
```

```
C:\>ssh -l admin 10.0.0.1
```

```
Open
```

```
Password:
```

```
R1#config t
```

```
Enter configuration commands, one per line. End with  
CNTL/Z.
```

```
R1(config)#
```

The last thing to check is that only SSHv2 is allowed for remote management access.

Close the SSH connection then attempt to Telnet to R1 from PC1.

```
R1#exit
```

```
[Connection to 10.0.0.1 closed by foreign host]
```

```
C:\>telnet 10.0.0.1
```

```
Trying 10.0.0.1 ...Open
```

```
User Access Verification
```

```
Username: admin
```

```
Password:
```

```
R1#
```

Telnet access is enabled. Only allow SSH access in the virtual terminal configuration.

```
R1(config)#line vty 0 15
```

```
R1(config-line)#transport input ssh
```

Close the existing session on PC1 then verify Telnet connections are now rejected.

```
R1#exit
```

```
[Connection to 10.0.0.1 closed by foreign host]  
C:\>telnet 10.0.0.1  
Trying 10.0.0.1 ...Open
```

```
[Connection to 10.0.0.1 closed by foreign host]
```

Check SSH is set to version 2.

```
R1#show run | include ssh  
ip ssh version 1  
transport input ssh  
transport input ssh
```

SSH is currently version 1. Attempt to set it to version 2.

```
R1(config)#ip ssh version 2  
Please create RSA keys (of at least 768 bits size) to  
enable SSH v2.
```

The RSA keys must have been created with the default 512 bits size. 768 bits is required for SSHv2.

Regenerate the RSA keys with a bit size of at least 768.

```
R1(config)#crypto key generate rsa  
% You already have RSA keys defined named R1.flackbox.com .  
% Do you really want to replace them? [yes/no]: yes  
The name for the keys will be: R1.flackbox.com  
Choose the size of the key modulus in the range of 360 to  
2048 for your  
General Purpose Keys. Choosing a key modulus greater than  
512 may take  
a few minutes.
```

```
How many bits in the modulus [512]: 768  
% Generating 768 bit RSA keys, keys will be non-  
exportable...[OK]
```

We are now able to set SSH to version 2.

```
R1(config)#ip ssh version 2
```

We have now completed all the required tasks.